

Computational Marine Ecological Modelling report: NPZDO model

This report should document the NPZDO model that you have developed during the first half of the course. The report should contain:

1. An illustration of the state variables and the processes that occur between them
2. Description of the governing equations and the different terms in the equations
3. Description of the parameters and their units
4. Description of the numerical scheme including boundary conditions
5. Show that the solution is converged and independent of the grid size.
6. Discussion of the factors limiting phytoplankton growth
7. Description (with suitable figures) of the steady-state solutions of the model under stable summer conditions with low and high nutrient loading. Explain why the solutions are different in the two situations.
8. A sensitivity analysis of selected parameters entering the model, with interpretation of why a change in the parameters leads to the observed effects
9. A description of the seasonal succession of the state variables
10. Link to your code on GitHub

The report should be in the form of a pdf document.